

EDUGENIE — TECHNOLOGY STACK

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BACKEND

- Python 3.10+
- FastAPI (REST API framework)
- Uvicorn (ASGI server)
- Jinja2 (HTML templating)
- Pydantic (request validation, implicit)

AI LAYER

- Google Gemini API
- Model: gemini-3.8-flash
- SDK: google-genai
- Used for: Q&A, explanations, summaries, quiz generation,
and learning path recommendations

FRONTEND

- HTML5
- CSS3 (custom styling, responsive layout)
- Vanilla JavaScript (fetch API for AJAX calls)
- No frameworks needed (keeps it lightweight)

ENVIRONMENT MANAGEMENT

- python-dotenv (reads GOOGLE_API_KEY from .env)
- Virtual environment (venv)

- requirements.txt (dependency list)

DEVELOPMENT TOOLS

- VS Code
- Git & GitHub
- Windows PowerShell terminal

WHY THESE CHOICES

FastAPI: Fast, modern, automatic API docs, async support

Gemini Flash: Fast responses, free tier, excellent at education tasks

Jinja2: Simple templating, no JS framework needed

Vanilla JavaScript: Avoids build steps and node_modules

No database: Stateless design — nothing to persist, privacy-friendly